



Influence of atmospheric moisture on the gas evolution tolerance of halide solid electrolytes

Takeshi Usami^{1,2} · Naoto Tanibata² · Hayami Takeda² · Masanobu Nakayama²

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Abstract

Much attention has been paid on research and development on solid electrolytes for all-solid-state Li batteries. Although halide solid electrolytes such as Li_3YCl_6 and Li_3InCl_6 are promising due to fast Li ion conductivity and oxidation-resistant against positive electrode, a better understanding of their reactivity with atmospheric H_2O is required for commercialization. In this study, the gas evolution tolerances of Li_3YCl_6 and Li_3InCl_6 were investigated. Temperature-programmed desorption mass spectrometry (TPD-MS) experiments at dew points below $-60\text{ }^\circ\text{C}$ and gas detector tube experiments at dew points of $-30\text{ }^\circ\text{C}$ both revealed significant differences in the H_2O and HCl evolution behavior of Li_3YCl_6 and Li_3InCl_6 . In TPD-MS, the onset temperature of HCl evolution for Li_3YCl_6 ($\sim 100\text{ }^\circ\text{C}$) was significantly lower than that for Li_3InCl_6 ($\sim 220\text{ }^\circ\text{C}$), indicating that Li_3InCl_6 solid electrolytes have superior gas evolution tolerance. This difference may be attributable to differences in the retention of H_2O derived from the material synthesis stage and from contact with the atmosphere during the measurements. In particular, based on first-principles calculations, the low-temperature HCl evolution observed in Li_3YCl_6 was ascribed to the partial replacement of Cl^- ions by OH^- ions upon contamination with trace H_2O . Because the heating and drying of solid electrolytes (including slurries) are inevitable processes during battery manufacturing, these findings can aid in the rational design of halide solid electrolytes for all-solid-state batteries.

Keywords Halide solid electrolyte · All-solid-state li battery · Gas evolution tolerance · HCl evolution · Temperature-programmed desorption mass spectrometry · First-principles calculations

Introduction

Recently, interest in carbon neutrality has promoted a reassessment of energy devices, motivating a shift from gasoline vehicles to battery electric vehicles (BEVs) [1]. For this purpose, all-solid-state lithium batteries have attracted significant attention owing to their high theoretical energy density, and the implementation of bipolar structures [2] and thicker films [3] has been considered. To utilize all-solid-state batteries as an energy source for BEVs, sulfide solid electrolytes (SEs) with lithium-ion conductivity comparable to or higher than that of organic electrolytes have been

developed to ensure a high output density [4–7]. Disadvantageously, sulfide SEs are unstable in atmospheric moisture, which leads to decreased lithium-ion conductivity and the evolution of toxic H_2S gas, thereby increasing process control costs [8–14]. In addition, sulfide SEs are unstable on oxide-based cathodes, resulting in a shorter cycle life owing to material degradation and lower power output owing to increased interfacial resistance between the SE and cathode active material [15–19]. To overcome these issues, halide SEs have become an attractive alternative. In 2018, Asano et al. reported that Li_3YCl_6 and Li_3YBr_6 have high ionic conductivities (0.51 and 1.7 mS/cm, respectively). In addition, all-solid-state batteries fabricated with these halide SEs and an oxide cathode active material exhibited stable charge–discharge cycling [20]. As conventional sulfide SEs do not have a sufficiently wide oxidation potential window, the oxide cathode must be coated with another material, such as LiNbO_3 . In contrast, halide SEs are thought to have inherent stability against oxidation [21, 22]. However, metal chlorides can undergo side reactions, such as reacting

✉ Masanobu Nakayama
masanobu@nitech.ac.jp

¹ Frontier Research and Development Division, Aichi Steel Corporation, Arai, Tokai, Aichi 476-8666, Japan

² Department of Advanced Ceramics, Nagoya Institute of Technology, Gokiso, Nagoya, Showa, Aichi 466-8555, Japan

with moisture to release HCl gas. In 2019, Li et al. reported that Li_3InCl_6 , which was synthesized in aqueous solution, showed high water resistance in atmospheric environments [23]. Subsequently, many studies have focused on improving the ionic conductivity of Li_3InCl_6 by doping with different elements [24–28]. Nevertheless, a precise understanding of the reactivity of halide SEs with H_2O is required for rational compositional or structural design. Based on in-depth experiments and observations, Wang et al. discussed the reaction mechanism responsible for the H_2O sensitivity of halide SEs [29]. They demonstrated that when Li_3InCl_6 comes into contact with atmospheric moisture, it becomes $\text{Li}_3\text{InCl}_6 \cdot 2\text{H}_2\text{O}$, which eventually deliquesces to form an acidic liquid. However, there have been very few reports on gas generation and thermal (exothermic/endothermic) behavior, even though it is essential for the production of halide SEs for commercial applications. Very recently, Chen et al. have reported chemical stability under ambient air and dry room conditions for chloride solid electrolytes, and confirmed irreversible decomposition of Li_3YCl_6 , while Li_3InCl_6 show less reactive against water and form stable hydrate which can be recovered to Li conductive Li_3InCl_6 phase by heating at 260 °C under vacuum [30].

In this study, the gas evolution behavior of typical halide SEs, Li_3YCl_6 and Li_3InCl_6 , during heating in a low-moisture and inert atmosphere was investigated using temperature-programmed desorption mass spectrometry (TPD-MS). In addition, the amount of HCl gas evolved was evaluated using a detector tube gas analyzer. The pH of the SEs was also investigated upon exposure to H_2O in air or in an atmosphere with a dew point of -30 °C. Furthermore, first-principles calculations were used to examine the stability of the SEs in the presence of moisture.

Methods

Experimental procedures

Synthesis and characterization of SEs

SEs were synthesized using mechanochemical methods. A stoichiometric mixture of LiCl (99.9%, Sigma-Aldrich, St. Louis, MO, USA) and YCl_3 (99.9%, Sigma-Aldrich, St. Louis, MO, USA) or InCl_3 (99.99%, High Purity Chemical Laboratory, Saitama, Japan) was added to a 45 mL ZrO_2 pot. The mixture was milled with 34 ZrO_2 balls (diameter = 8 mm) using a planetary ball mill (P-7 classic-line, Fritsch Japan K.K., Tokyo, Japan) at 500 rpm for 1 h. and then allowed to rest for 15 min. For Li_3InCl_6 , the ball-milled sample was subsequently heat treated at 300 °C for 3 h to enhance Li ion conductivity according to literature [24].

The synthesized samples were characterized using X-ray diffraction (XRD; MiniFlex 600, Rigaku Corp., Tokyo, Japan). The ionic conductivity was measured at 298 K using the AC impedance method (VSP electrochemical analyzer, BioLogic, Grenoble, France) at an applied voltage of 10 mV and frequencies of 10^2 – 10^7 Hz. For these measurements, pellets (diameter = 10 mm, thickness = ~0.50 mm) were prepared by sandwiching the SE powder between stainless steel pins and compressing at 509 MPa. All procedures were performed under a dry Ar gas atmosphere maintained at a dew point of less than -80 °C.

TPD-MS measurements

The gas evolution behavior of the synthesized SEs was evaluated using TPD-MS. For TPD-MS, an MS instrument (GC/MS QP2010Plus, Shimadzu Corporation, Kyoto, Japan) was directly connected to a heating device with a temperature controller. The concentration change at the mass number corresponding to each gas generated by the sample during heating was calculated as a function of temperature and time (Fig. S1). For 15 min before measurement, dry He (purity 99.999%, dew point < -70 °C) was used as the carrier gas at a flow rate of 50 mL/min was used to displace the gas inside the apparatus. The same flow rate was used during the measurement to maintain a low-moisture atmosphere with a dew point of less than -60 °C. During measurement, the temperature was increased from room temperature to 500 °C at a rate of 10 °C/min.

FT-IR measurements

The vibrational states of water molecules in solid-state samples were investigated using attenuated total reflection Fourier transform infrared spectroscopy (ATR-FTIR, VERTEX 70v, Bruker Corporation, Billerica, Massachusetts, USA), where incident angle is set as 45 degrees. Samples are sealed in air-tight containers in a dry Ar gas atmosphere at a dew point of less than -80 °C, and transferred to the analyzer. The measurements were performed under dry N_2 atmosphere immediately after opening the containers in the sample chamber of the instrument.

Detector tube gas analyzer measurements

The gas evolution behavior of the synthesized SEs was also investigated using a detector tube gas analyzer. The synthesized SEs were sealed in a dry Ar gas atmosphere at a dew point of less than -80 °C and then moved to an Ar gas atmosphere at a dew point of -30 °C and placed in a beaker (Fig. S2). Subsequently, the sample was heated to a predetermined temperature on a hot plate and maintained at

that temperature for 10 min. The amount of evolved gas was determined using a detector tube gas analyzer (GC-100 S, GASTEC, Ayase, Japan) and a gas detector tube (No 14 L, GASTEC). A sealed environment was maintained to prevent the gas from dispersing during heating and standing. The amount of gas was calculated from the concentration measured using the gas detector tube and the volume of the glass container, and the results were reported as the amount of gas produced per gram based on the sample weight.

Static test of SE powder using pH test paper

The synthesized SEs were sealed in a dry Ar gas atmosphere at a dew point of less than $-80\text{ }^{\circ}\text{C}$. A portion of the SE powder SE sample was transferred to an Ar gas atmosphere at a dew point of $-30\text{ }^{\circ}\text{C}$ or to air (45% humidity at $25\text{ }^{\circ}\text{C}$) and placed on pH test paper (universal indicator paper pH 1–14, MACHEREY-NAGEL GmbH & Co. KG, Düren, Germany). In addition, to confirm the color corresponding to each pH value, the pH test papers were dipped in aqueous solutions adjusted to various pH values using ion-exchanged water, HCl, and NaOH.

The remaining portion of the SE powder was placed in an atmosphere at a dew point of $-30\text{ }^{\circ}\text{C}$ for 60 min, and the products were characterized by XRD.

Calculation procedures

Density functional theory (DFT) calculations were performed to investigate the reaction of the halide SEs with H_2O . Structure models for Li_3InCl_6 and Li_3YCl_6 were referred to the inorganic crystal structure database (ICSD), where ID numbers of Li_3InCl_6 and Li_3YCl_6 are 17,638 and 29,969, respectively. Li_3InCl_6 belongs to monoclinic (C2/m), while Li_3YCl_6 belongs to trigonal (P-3 m). Superstructure models

for Li_3InCl_6 and Li_3YCl_6 , corresponding to $\text{Li}_{48}\text{In}_{16}\text{Cl}_{96}$ and $\text{Li}_{72}\text{Y}_{24}\text{Cl}_{144}$, respectively, were prepared in accordance with a previous study [28]. This study investigated the diffusivity of Li in Li_3InCl_6 and Nb- and/or Zr-doped compounds.

The partial replacement of Cl^- ions by OH^- ions in $\text{Li}_{48}\text{In}_{16}\text{Cl}_{96}$ and $\text{Li}_{72}\text{Y}_{24}\text{Cl}_{96}$ was considered to model the reaction with H_2O (as described later). After structural optimization, the total energy was evaluated using the Vienna ab initio software package (VASP) [30–34] with the projector augmented-wave method [35, 36] and a plane-wave basis set. The generalized gradient approximate exchange correlation function developed by Perdew, Burke, and Ernzerhof and modified for solid materials (PBEsol-GGA) was used [37]. The cut-off energy of the plane-wave basis was set to 500 eV. A $2 \times 3 \times 1$ Monkhorst–Pack k-point grid was used for optimization calculations.

Results and discussion

Experimental results

The XRD patterns and AC impedance plots of Li_3InCl_6 and Li_3YCl_6 under dry conditions are shown in Fig. 1(a) and (b), respectively. All measurements were performed in a dry room (dew point $< -30\text{ }^{\circ}\text{C}$) immediately after transfer from the dry box (dew point $< -30\text{ }^{\circ}\text{C}$). Both the XRD patterns and AC impedance plots of the Li_3InCl_6 and Li_3YCl_6 samples agree well with those in previous studies [20, 28].

Figure 2(a) and (b) show the TPD-MS results for H_2O ($m/z=18$) and HCl ($m/z=36$), respectively. In addition, for comparison purposes, the profiles for both gases are shown for Li_3YCl_6 and Li_3InCl_6 in Fig. 2(c) and (d), respectively. Cl_2 gas was not detected in the TPD-MS measurements. The total molar ratio of H_2O to the theoretical molar

Fig. 1 Characterization of the synthesized SE samples. (a) XRD patterns of SEs. Li_3YCl_6 was synthesized in a planetary ball mill and Li_3InCl_6 was synthesized by heating at $300\text{ }^{\circ}\text{C}$ for 3 h after treatment in a planetary ball mill. (b) Cole–Cole plots of SEs (black: Li_3YCl_6 , red: Li_3InCl_6)

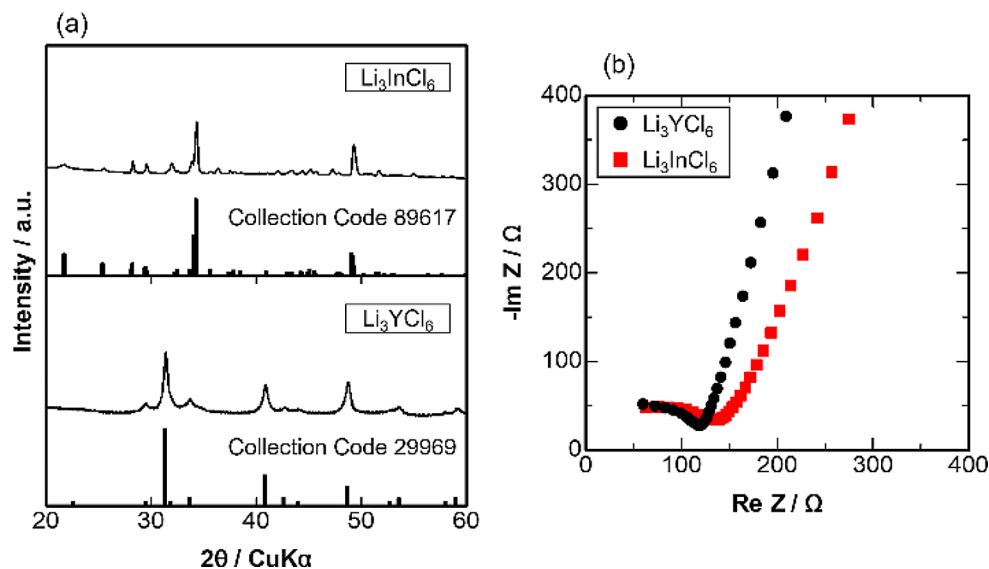
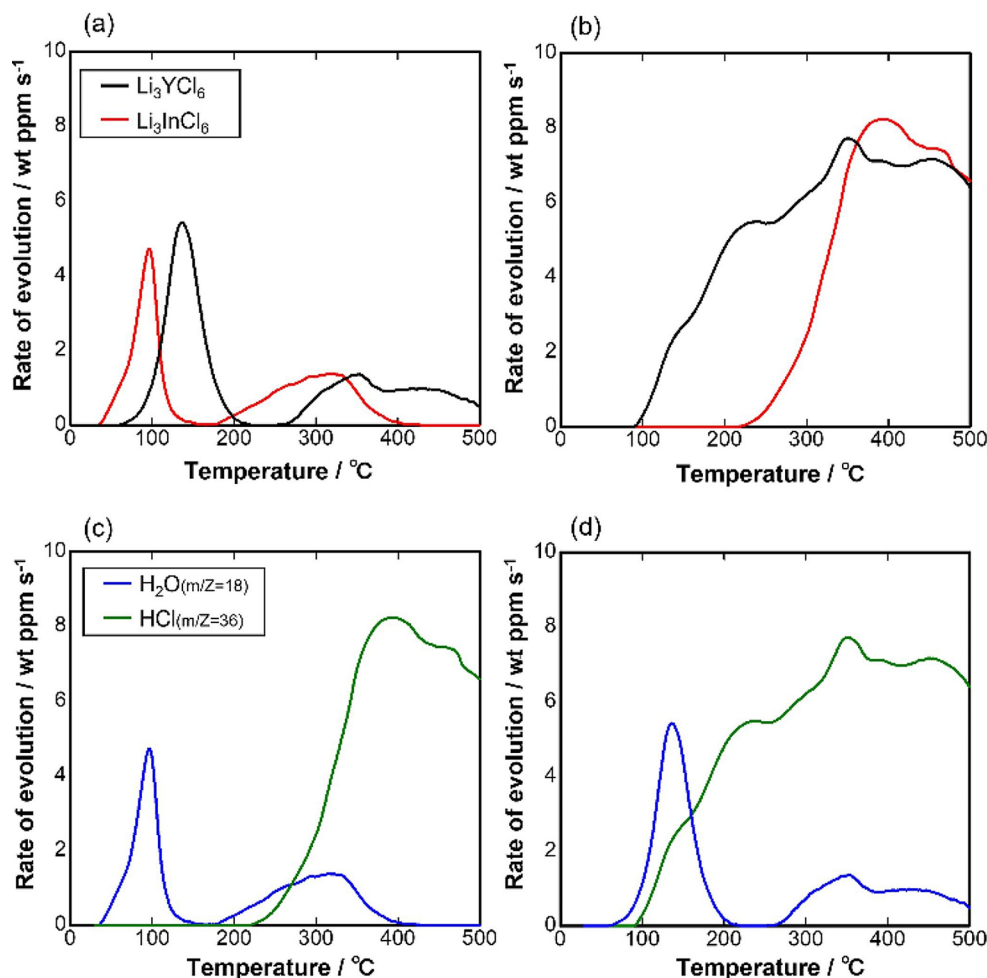
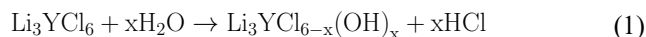


Fig. 2 TPD-MS results (temperature vs. gas generation rate curve) for the synthesized SEs. **(a)** and **(b)** Rates of evolution for H₂O ($m/z = 18$) and HCl ($m/z = 36$), respectively (black: Li₃YCl₆, red: Li₃InCl₆). **(c)** and **(d)** Rates of evolution for Li₃InCl₆ and Li₃YCl₆, respectively (blue: H₂O ($m/z = 18$), green: HCl ($m/z = 36$))



amount of Cl in the SEs was 3.1 and 5.0 mol% for Li₃InCl₆ and Li₃YCl₆, respectively (Fig. 2(a)), suggesting that the SE samples either retained moisture during synthesis or adsorbed moisture from the TPD-MS measurement environment at a dew point below -60 °C. The TPD-MS peak for H₂O evolution from Li₃InCl₆ appeared at ~ 100 °C, equivalent to the boiling point of water. In contrast, for Li₃YCl₆, this peak appeared at a higher temperature (~ 140 °C). This difference may indicate differences in the chemical interactions between the SEs and trace H₂O originating from the synthesis and/or analysis environments, even in a humidity-controlled atmosphere. We infer that H₂O molecules are physically adsorbed on the surface of Li₃InCl₆ because of the agreement between the TPD-MS peak and the boiling point of H₂O. In contrast, H₂O penetrates the interior of the Li₃YCl₆ particles, resulting in H₂O evolution at higher temperatures. The onset temperature for HCl evolution was much higher for Li₃InCl₆ (~ 220 °C) than for Li₃YCl₆ (~ 100 °C) (Fig. 2(b)). For Li₃InCl₆, no correlation is observed between the gas evolution behavior of H₂O and HCl. However, for Li₃YCl₆, the temperatures of the H₂O peak and HCl shoulder agree well (~ 140 °C). Hence, we

assume that the HCl evolution reaction is triggered by H₂O gas, as shown in Eq. (1):



In this reaction, the Cl⁻ ions are partially replaced by OH⁻ ions. Equation (1) is also supported by the ATR-FTIR spectra shown in Fig. 3. For Li₃YCl₆ (Fig. 3(a) and (b)), the peaks derived from hydroxyl group are broad. In contrast, the corresponding peaks for Li₃InCl₆ (Fig. 3(c) and (d)) are relatively sharp, which suggests that the H₂O molecules are fixed in the lattice (i.e., crystalline water). Notably, a relatively strong peak appears at approximately 1600 cm^{-1} before the exposure test for Li₃InCl₆, indicating that a trace amount of H₂O, similar to crystalline water, is retained during the synthesis stage (dew point less than -80 °C). A schematic of the processes occurring at relatively low temperatures is shown in Fig. 4. Note that above H₂O-involved reaction was conducted in humidity-controlled environments, where samples were kept in a glove box at the dew point of dry Ar (< -80 °C), whereas the TPD-MS measurements were conducted in a dry He environment. Nevertheless, trace

Fig. 3 ATR-FTIR spectra of (a, b) Li_3YCl_6 and (c, d) Li_3InCl_6 . The peaks attributed to OH vibrations are shown, where γ - and δ -OH indicate stretching and bending vibrations, respectively. The black and red lines correspond to the samples before and after exposure to an atmosphere with a dew point of -30°C for 10 min

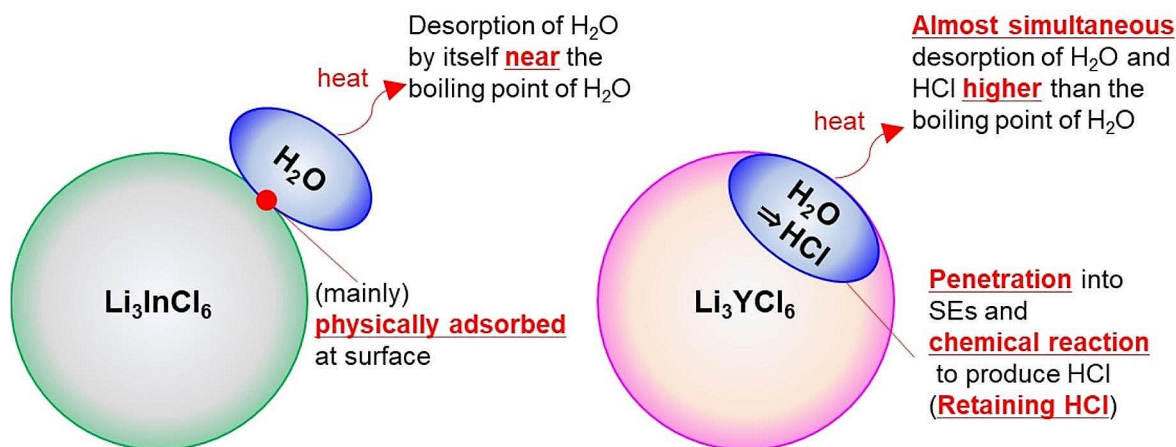
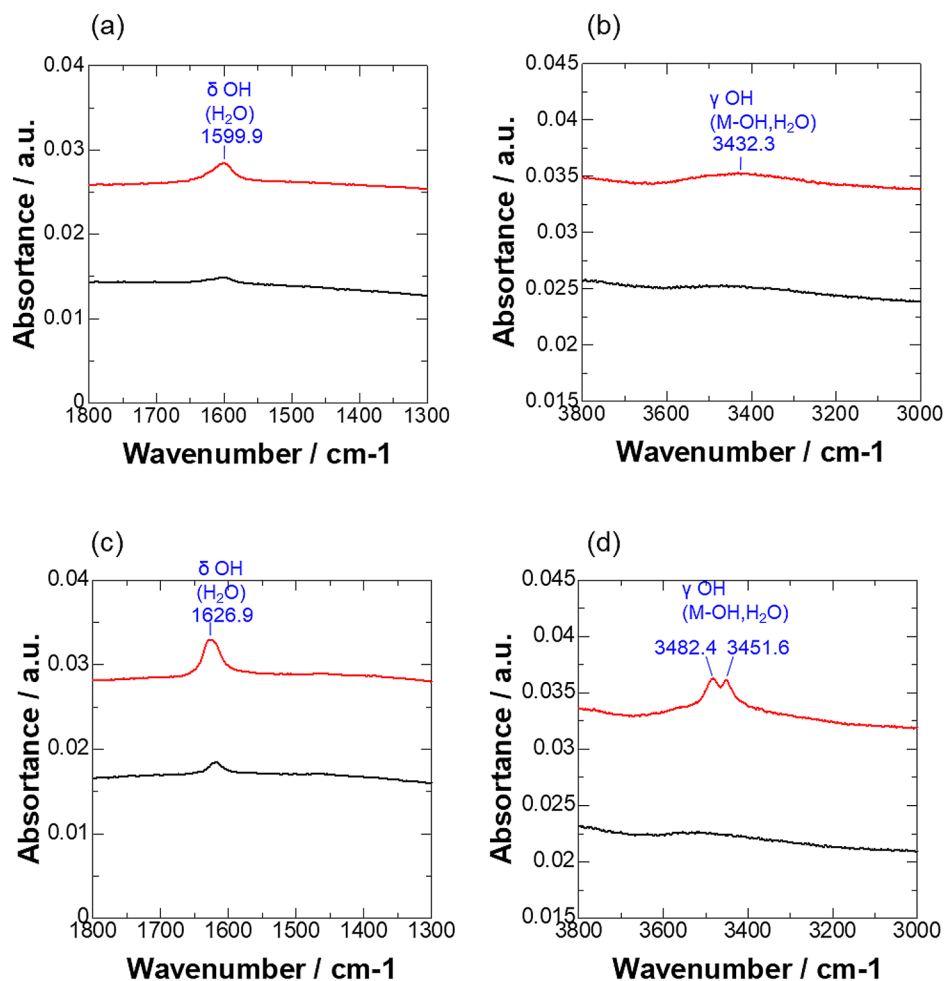


Fig. 4 Schematic diagram showing the form in which the solid electrolyte retains moisture from the atmosphere

H_2O molecules may affect the generation of HCl gas from Li_3YCl_6 at $\sim 140^\circ\text{C}$. Furthermore, the lack of strong interactions between Li_3InCl_6 and H_2O indicates that the HCl evolution reaction with H_2O can be controlled by selecting appropriate metal ions for Li-containing chlorides.

Figure 5 shows the amount HCl gas evolved during detector tube gas analyzer measurements at a dew point of -30°C as a function of temperature. This method can specify the evolution of HCl gas, sensitively and inexpensively. Similar to the TPD-MS results (Fig. 2), HCl gas evolution

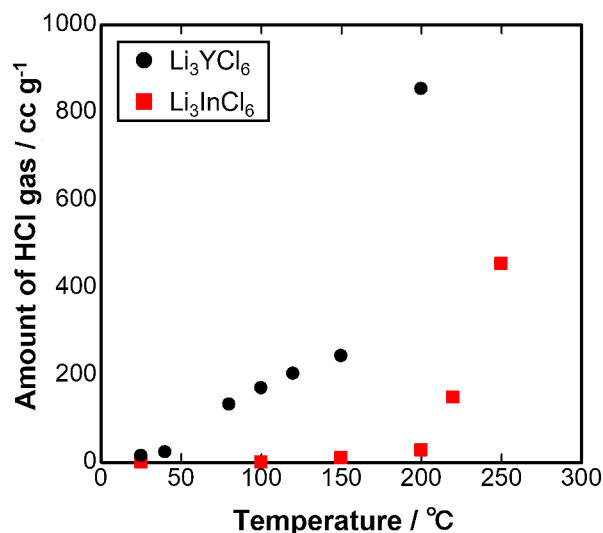


Fig. 5 Amount of generated HCl, as measured using a detector tube gas analyzer at a dew point of $-30\text{ }^{\circ}\text{C}$ (black: Li_3YCl_6 , red: Li_3InCl_6)

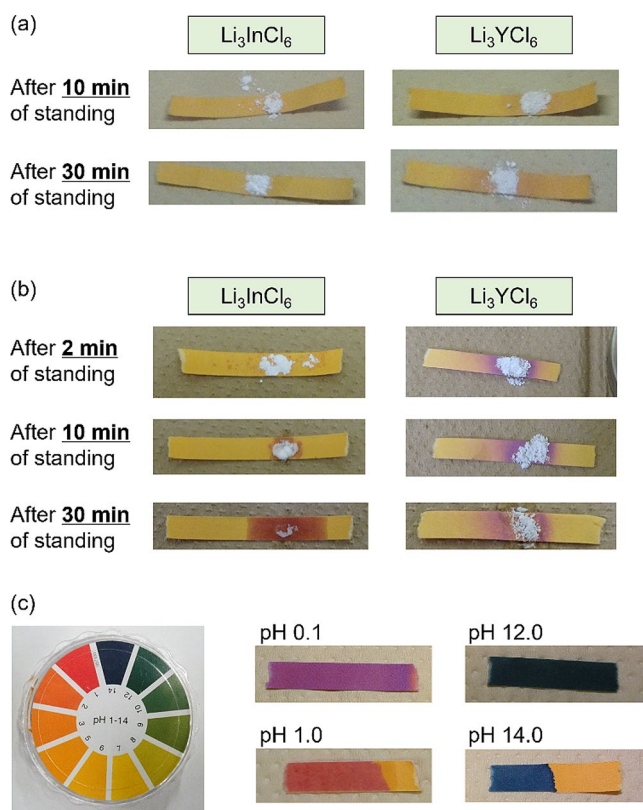


Fig. 6 Photographs of pH test papers after exposure to the SEs for pre-determined times. (a) and (b) Static exposure to an atmosphere with a dew point of $-30\text{ }^{\circ}\text{C}$ and atmospheric conditions, respectively. (c) Color change of the pH test paper dipped in aqueous solutions with various pH values

from Li_3YCl_6 was detected at a lower temperature than that from Li_3InCl_6 . The onset temperature of HCl gas evolution for Li_3YCl_6 was below $100\text{ }^{\circ}\text{C}$, and a trace amount of HCl gas was detected at room temperature. This behavior is consistent with the hypothesis that H_2O penetrates the SE powder particles, even in a humidity-controlled atmosphere. The observed onset temperature for HCl evolution is lower with the detector tube gas analyzer than with TPD-MS because of the different operating atmospheres (dew points of -30 and $-60\text{ }^{\circ}\text{C}$, respectively). However, with Li_3InCl_6 , HCl gas was observed at temperature above $150\text{ }^{\circ}\text{C}$, which is slightly lower than the temperature obtained using TPD-MS. Note that the same results are observed for Li_3YCl_6 heated at $300\text{ }^{\circ}\text{C}$ for 3 h as shown in Figure S3 in the Supporting Information, so that the HCl gas evolution behavior does not change the heat treatment after ball milling. This behavior suggests that increasing the ambient moisture content not only causes water to be retained through surface adsorption but also promotes internal chemical reactions, as in Li_3YCl_6 , HCl gas evolution above $200\text{ }^{\circ}\text{C}$ occurred with both Li_3YCl_6 and Li_3InCl_6 , as also observed in the TPD-MS measurements. Wang et al. proposed that decomposition to InCl_3 and LiCl and subsequent hydrolysis reaction leads to the formation of HCl [29]. We infer that a similar hydrolysis reaction mechanism proceeds in Li_3YCl_6 at temperatures above $200\text{ }^{\circ}\text{C}$.

Figure 6(a) shows photographs of the Li_3YCl_6 and Li_3InCl_6 powders placed on pH test paper in a dry atmosphere (dew point of $-30\text{ }^{\circ}\text{C}$) after 30 min as well as the colors of the pH test paper when dipped in aqueous solutions with various pH values. After 30 min, the pH test paper with Li_3InCl_6 showed little coloration, whereas that with Li_3YCl_6 showed a color change corresponding to acidic conditions, probably due to HCl gas evolution. These results are consistent with trace HCl gas being evolved without heating, as shown by the detector tube gas analyzer test (Fig. 5). To clarify the different gas evolution behavior of Li_3InCl_6 and Li_3YCl_6 , the same test was conducted in an ambient atmosphere ($25\text{ }^{\circ}\text{C}$ and 45% relative humidity), as shown in Fig. 6(b). Li_3InCl_6 exhibited deliquescence, absorbing H_2O from the atmospheric moisture and dissolving, and color change of the pH test paper indicated acidity, in good agreement with the system reported by Wang et al. [26]. In contrast, Li_3YCl_6 showed no deliquescence in the early stage of the reaction (~ 2 min); however, the pH test paper exhibited an acidic color change, which gradually spread without dissolution of the Li_3YCl_6 powder, even after 30 min. XRD analysis of the Li_3YCl_6 powder after 30 min in the ambient atmosphere showed peaks corresponding to Li_3YCl_6 and LiCl (Fig. 7). These results indicate that the initial reaction with water occurs without disrupting the matrix structure. We infer that Li_3InCl_6 dissolves easily in H_2O when

Fig. 7 XRD patterns after exposure of the SE samples to an atmosphere with a dew point of $-30\text{ }^{\circ}\text{C}$ for up to 60 min. Panels (a) and (b) show the results for Li_3InCl_6 and Li_3YCl_6 , respectively

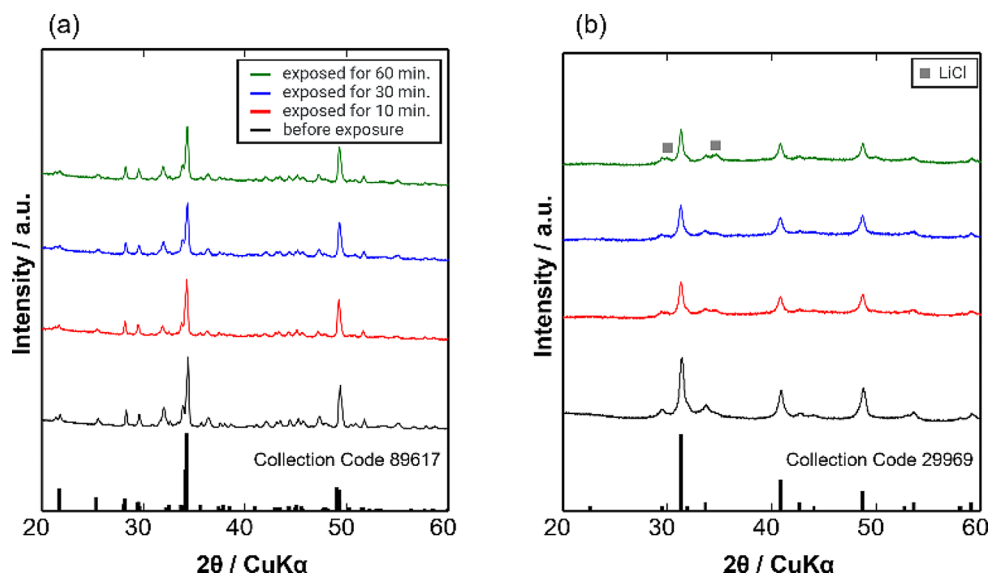


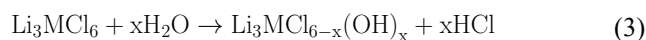
Table 1 Defect formation energies (ΔE_{def}) values calculated for Eq. (3). The values in brackets represent energies after zero-point energy correction

Reaction	Li_3YCl_6	Li_3InCl_6
Equation (3) (eV / H_2O)	0.63 (0.24)	2.94 (2.55)

sufficient H_2O molecules are present for solvation, whereas the partial replacement of Cl^- by OH^- ions is unlikely to proceed with a small amount of H_2O . The opposite behavior occurs for Li_3YCl_6 .

Calculation results

Based on the experimental observations, we considered two types of reactions between Li_3MCl_6 and trace H_2O during the early stages of decomposition, where the host structure of Li_3MCl_6 is maintained.



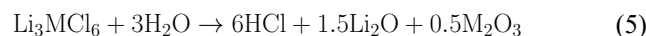
In Eq. (2), Li^+ is replaced with H^+ , which results in the formation of alkaline LiOH . In contrast, in Eq. (3), Cl^- is replaced with OH^- , leading to the formation of acidic HCl . The experimental observation of HCl gas evolution suggests that Eq. (3) is likely to proceed. Therefore, defect formation energies, ΔE_{def} , for Eq. (3), where $\text{M} = \text{In}$ and Y , were evaluated using DFT calculations (Table 1). Zero-point energy (ZPE) corrections are also considered by referring Materials Project datasets (molecule IDs, e67f3- H_2O 1-0-1 and b3c0- ClH 1-0-1 for H_2O and HCl , respectively) [38], where ZPE for H_2O and HCl are 0.579 and 0.185 eV, respectively. Positive ΔE_{def} values were observed for both Li_3InCl_6 and Li_3YCl_6 because of the absolute zero-Kelvin assumption in

Table 2 Decomposition reaction energy with H_2O (ΔE_{decomp}) values calculated for Eqs. (4)–(6). The values in brackets represent energies after zero-point energy correction

Reaction	Li_3YCl_6	Li_3InCl_6
Equation (4) (eV / H_2O)	0.76 (0.55)	2.49 (2.28)
Equation (5) (eV / H_2O)	1.87 (1.66)	2.03 (1.82)
Equation (6) (eV / H_2O)	0.86 (0.55)	1.01 (0.71)

this calculation and because the free energies of the gas-phase H_2O and HCl molecules are largely reduced by the entropic effect. Hence, HCl gas evolution is more likely to occur from Li_3YCl_6 than from Li_3InCl_6 , in agreement with the TPD-MS results at lower temperatures ($< 200\text{ }^{\circ}\text{C}$), as shown in Fig. 2.

We also investigated the decomposition reaction energy with H_2O , ΔE_{decomp} , which accompanies reconfiguration of the host structure of Li_3MCl_6 . The following three reactions were considered:



The calculated values for the decomposition reaction energy per mole of H_2O for reactions in Eqs. (4)–(6) are listed in Table 2. The decomposition energy for the reaction in Eq. (5) was high for both $\text{M} = \text{Y}$ and $\text{M} = \text{In}$, indicating that the reaction is unlikely to proceed. This suggests that the formation of Li_2O does not occur as readily as that of LiCl or LiOH . In contrast, the decomposition energy for the reactions in Eq. (4) and Eq. (6) when $\text{M} = \text{Y}$ and $\text{M} = \text{In}$, respectively, was below 1 eV/ H_2O , which is comparable to the decomposition reaction energy value for the reaction in

Eq. (3). The reaction in Eq. (3) involves the formation of a solid solution phase, $\text{Li}_3\text{MCl}_{6-x}(\text{OH})_x$, without a significant change of the host structure. In contrast, reactions in Eq. (4) and Eq. (6) involve structural changes requiring the diffusion of all ions, making them kinetically slower than that in Eq. (3). Therefore, the reaction in Eq. (3) is dominant at low temperatures, resulting in the observation of HCl formation only for $\text{M}=\text{Y}$, which has a lower decomposition energy (Table 1). In contrast, the decomposition reactions in Eq. (4) for $\text{M}=\text{Y}$ and Eq. (6) for $\text{M}=\text{In}$ are expected to proceed at higher temperature. Thus, the HCl release reactions observed above 200 °C in the TPD-MS profile (Fig. 2) are considered to correspond to the reactions in Eqs. (4) and (6).

Conclusions

To promote the widespread use of all-solid-state batteries, it is important to design safe processes and properties, such as ion conductivity and formability. In this study, we focused on gas evolution resulting from chemical reactions between halide SEs and H_2O in the atmosphere. Li_3InCl_6 and Li_3YCl_6 exhibited different interactions with H_2O molecules and reaction mechanisms for HCl gas formation. Li_3InCl_6 was relatively resistant to HCl gas evolution because a small amount of H_2O was physically adsorbed on the surface and the defect formation energy for Eq. (3) was higher. In contrast, Li_3YCl_6 underwent defect formation at lower temperatures because H_2O was adsorbed within the bulk and the defect formation energy was lower. At high temperatures or in atmospheres containing more H_2O , decomposition proceeded for both Li_3InCl_6 and Li_3YCl_6 . Though the knowledge on water tolerance for chloride electrolytes is limited to Li_3InCl_6 and Li_3YCl_6 in this study, the present study unveiled the reaction mechanism between chloride electrolytes and water. It is thought that penetration of water into the bulk of particles, leading to replacement of Cl^- by OH^- , is crucial for HCl gas release at low temperature. This phenomenon can be accelerated or suppressed, depending on the amount of H_2O in the atmosphere, the metal ion or crystal structure in the Li_3MCl_6 composition.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10008-024-05880-z>.

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Author contributions T.U. and Masanobu Nakayama conceived and directed the project. T.U. performed experimental and computational evaluations. T.U. and M.N. analyzed output data. The manuscript was mainly written by T.U. and M.N. through the contributions of all authors. All the authors have approved the final version of the manuscript.

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Declarations

Competing interests The authors have no competing interest to declare that are relevant to the content of this article.

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